

Gaurav Rai

Ph.D. Research Scholar, IIIT-Delhi

✉ gauravr@iiitd.ac.in | <https://gauravraiitd.github.io/> | [gauravraiIIITD](#)

AREAS OF RESEARCH

Computer Graphics: Skeleton Deformation, Mesh Processing, Rigging, Differentiable Rendering

Generative AI: Text-to-Video Synthesis, Unsupervised Animation, Autoregressive Transformers

Computer Vision: Sketch-to-Video Synthesis, Motion Estimation & Point Tracking, Scene Understanding

EDUCATION

Indraprastha Institute of Information Technology, Delhi, India *Aug 2020-2026 (Thesis submitted)*

PhD Research Scholar, Computer Science and Engineering

CGPA: 8.24/10.0

Advisor: [Dr. Ojaswa Sharma](#)

Gurukula Kangri Vishwavidyalaya, Haridwar, Uttarakhand

2016 - 2020

B.Tech. Computer Science and Engineering

CGPA: 8.52/10.0

PUBLICATIONS

PUBLISHED

[1] **Space-aware Indoor Furniture Recommendation and Placement using Agentic AI**

Prashant Goswami, **Gaurav Rai***, V S A Akula*, Ojaswa Sharma, Muhammad Imran

2025

[IEEE International Conference on Intelligent Reality \(ICIR 2025\)](#)

Project Page: <https://graphics-research-group.github.io/SAIFRaP/>

In collaboration with [IKEA](#).

[2] **LiveImage: Motion condition guided diffusion model for video motion transfer**

Gaurav Rai, Ojaswa Sharma

2025

[International Conference on Multimedia & Expo \(ICME\)](#)

[3] **Text-to-Video Diffusion Models with Temporal Consistency and Rigidity Constraints**

Gaurav Rai, Ojaswa Sharma

2025

[20th International Conference on Computer Graphics Theory and Applications \(GRAPP\)](#)

Project Page: <https://graphics-research-group.github.io/ESA/>

[4] **SKETCHANIM: Real-time Sketch Animation Transfer from Videos**

Gaurav Rai, Shreyas Gupta, and Ojaswa Sharma

2024

[ACM SIGGRAPH / EUROGRAPHICS Symposium on Computer Animation \(SCA\)](#)

Published in [Computer Graphics Forum \(CGF\)](#)

Project Page: <https://graphics-research-group.github.io/SketchAnim/>

UNDER REVIEW

[5] Generative AI for Video-driven Image Animation: A Survey

Gaurav Rai, Akshit Salhotra, Ojaswa Sharma

2026

[6] Text-driven 2D sketch to 3D Animation via Diffusion-Guided Skeleton Optimization

Gaurav Rai, Ojaswa Sharma

2026

[7] A Survey on Sketch Animation Methods

Gaurav Rai, Ojaswa Sharma

2026

PROJECTS

TexSketch | *Pytorch, OpenCV*

Jan 2026 – Present

- TexSketch: A Large-Scale Textured Sketch Dataset Generated from Line Art
- We provide a benchmark dataset of 16,000 sketch–texture sample pairs.

Sketch2Motion | *Pytorch, triangle, OpenCV, Scikit-learn*

April 2025 – March 2026

- Sketch2Motion: Text-driven 2D sketch to 3D Animation via Diffusion-Guided Skeleton Optimization
- Given a hand-drawn sketch, we create a 3D model of the 2D sketch and animate the 3D sketch model using text prompt.

Indoor furniture placement | *Pytorch, Pytorch3d, OpenCV, Scikit-learn, OpenGL, Blender*

Sept 2024 – Jul 2025

- Space-aware Indoor Furniture Recommendation and Placement using Agentic AI
- In collaboration with [IKEA](#).
- Given a floor plan, our proposed method performs space-aware and object-aware scene synthesis for automatic indoor furniture recommendation and placement.

Enhancing Sketch Animation | *Pytorch, libigl, OpenCV, Scikit-learn*

Feb 2024 – Oct 2024

- Text-to-Video Diffusion Models with Temporal Consistency and Rigidity Constraints
- Given a static sketch and text prompt describing the motion, our method animates the sketch as per the input text description. The various valuable applications are children’s ebooks, entertainment videos, video editing, etc.

LiveImage | *Pytorch, OpenCV, Scikit-learn*

Aug 2022 – Dec 2024

- Motion condition guided diffusion model for video motion transfer
- An approach for image animation that has exciting applications in various visual tasks such as video editing and animation. It takes an image and a driving video as input and generates the animation video of the image based on the driving video’s motion.

SketchAnim | *Pytorch, Pytorch3d, open3d, libigl, OpenCV, Scikit-learn*

Aug 2021 – Dec 2023

- Real-time sketch animation transfer from videos
- Allows novice users to bring static drawings to life by applying deformation-based animation effects extracted from video examples.

EXPERIENCES

- Research Associate, Center for Design and New Media, IIIT Delhi** *Sept 2025 - Jan 2026*
- Visiting Researcher, BTH Sweden** *Sept 2024 - Feb 2025*
- Teaching Assistant, Advance Visualization** *Feb 2026*
- PG Diploma in Data Science in Health and Climate Change for Social Impact, IIIT-Delhi
- Teaching Assistant, IIIT-Delhi** *Aug 2020 - May 2024*
- Courses:** Computer Graphics [CSE333/533] (Monsoon 23, 25), Scientific Computing [MTH373/573] (Monsoon 22), Linear Algebra [MTH100] (Winter 21)
- Teaching Assistance** at [ACM Summer School on Shape Modelling](#) Topic: Introduction to libigl at IIIT Delhi (2022).

ACHIEVEMENTS

- Received ACM Student Travel Grant for [ICME 2025](#).
- Overseas Research Fellowship for a research visit to Blekinge Institute of Technology, Karlskrona, Sweden. Support of USD 12k for six months. (Sept 2024 - Feb 2025)
- Awarded the prestigious CHANAKYA Doctoral Fellowship 2024-2025.
- Reviewer at IEEE [ICME 2024](#), [ICME 2025](#), and [CVIP 2025](#).
- Qualified Graduate Aptitude Test in Engineering ([GATE](#)) in Computer Science and Engineering (2020).

TALKS AND PRESENTATIONS

- Presented the LiveImage paper at [ICME 2025](#) in Nantes, France.
- Presented the Enhancing Sketch Animation paper at [VISIGRAPP 2025](#) in Porto, Portugal.

SKILLS

Programming Languages: C/C++, Python.

Tools: OpenGL, Qt, Libigl, CGAL, Pytorch, OpenCV, Scikit-learn.

Research Interest: Computer Graphics, Generative AI, Visual Computing, and Computer Vision.

REFERENCES

- Dr. Ojaswa Sharma:** Associate Professor, Department of Computer Science, IIIT Delhi, India [ojaswa@iiitd.ac.in]
- Dr. Jainendra Shukla:** Associate Professor, Department of Computer Science, IIIT Delhi, India [jainendra@iiitd.ac.in]
- Prof. A V Subramanyam:** Professor, Department of Computer Science, IIIT Delhi, India [subramanyam@iiitd.ac.in]